

EasyShield v2.5: A Lightweight Real-Time Anti-Spoofing System for Edge Devices

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Abstract This paper presents *EasyShield v2.5*, a lightweight face presentation attack detection (PAD) system optimized for realtime deployment on embedded platforms. The proposed architecture combines YOLOv12n’s efficient feature extraction with a streamlined binary classification head, enabling fast, accurate spoof detection from RGB input. A mirrored face pair training strategy is adopted to improve generalization, enforcing liveness-specific feature learning. EasyShield is trained on a curated 12,000-image dataset targeting 2D attacks (print and replay), with MTCNN-based face extraction and spoof-aware augmentation. On a held-out test split, EasyShield achieves 99.82% in-domain accuracy. Under a cross-domain setting using the NUAA and Tapakah datasets—without fine-tuning—it achieves 92.3% accuracy, 98.61% AUC, and 6.25% Equal Error Rate (EER), outperforming state-of-the-art lightweight PAD baselines. The system runs at 75 ms per frame on a Raspberry Pi 4 (1 GB RAM) without GPU acceleration. With a $3\times$ reduction in FLOPs and more than $2\times$ fewer parameters compared to competitors, EasyShield v2.5 delivers a secure, interpretable, and deployable PAD solution for edge-based biometric systems aligned with Zero Trust Architecture (ZTA) principles.

Keywords Face anti-spoofing · YOLOv12n · Embedded image processing · Real-time PAD · Edge AI · Biometric security · Zero Trust

List of Acronyms

- **PA** – Presentation Attack
- **PAD** – Presentation Attack Detection

- **FAS** – Face Anti-Spoofing
- **AUC** – Area Under the Curve
- **EER** – Equal Error Rate
- **APCER** – Attack Presentation Classification Error Rate
- **BPCER** – Bona Fide Presentation Classification Error Rate
- **ACER** – Average Classification Error Rate
- **FPS** – Frames Per Second
- **YOLO** – You Only Look Once
- **CNN** – Convolutional Neural Network
- **RNN** – Recurrent Neural Network
- **ZTA** – Zero Trust Architecture
- **TPR** – True Positive Rate
- **FPR** – False Positive Rate
- **ROC** – Receiver Operating Characteristic
- **PR** – Precision-Recall
- **MTCNN** – Multi-task Cascaded Convolutional Network

1 Introduction

Facial biometrics are central to modern authentication, securing access to mobile applications, digital banking, and physical systems [2]. Yet they remain vulnerable to *presentation attacks* (PAs), where adversaries spoof recognition with printed photos or replayed videos (2D attacks) [21]. Without liveness detection, even advanced recognition systems can be bypassed. *Presentation attack detection* (PAD), or *face anti-spoofing* (FAS), addresses this by distinguishing genuine inputs from spoofed ones [3].

The spread of edge-based devices, such as smartphones and embedded cameras, demands FAS solutions that are compact, real-time, and resource-efficient. To

meet this need, we present EasyShield v2.5, a lightweight framework for RGB-based liveness detection on constrained edge hardware. The system is optimized for real-time operation and robust generalization to unseen attack types, validated through cross-domain tests on public and custom datasets and deployment trials on embedded hardware.

The major contributions of this work are fourfold:

- A compact FAS architecture coupling a pruned YOLO-v12n backbone with a binary classification head, enabling efficient spoof detection on low-power devices.
- A mirrored face pair training strategy promoting spoof-specific feature learning and reducing identity bias, implemented in a modular dataset pipeline.
- A curated dataset of 12,000 RGB face images supporting supervised spoof detection across varied conditions and attack types.
- Extensive evaluation on public benchmarks (OULU-NPU, SiW) and the proposed dataset, demonstrating competitive cross-domain performance and low-latency deployment on a Raspberry Pi 4 within edge-oriented and Zero Trust constraints.

EasyShield’s standalone design supports integration into Zero Trust Architecture (ZTA)-compliant biometric systems, particularly in constrained or privacy-sensitive settings.

The rest of the study is organized as follows:

2 Related Work

FAS has evolved from handcrafted features to deep learning models, and more recently to lightweight designs for real-time deployment. This section reviews these three phases to position the novelty of EasyShield v2.5.

2.1 Traditional Handcrafted Approaches

Early FAS methods relied on features such as Local Binary Patterns (LBP), Histogram of Oriented Gradients (HOG), and color descriptors [16, 5]. Temporal cues like blinking and head movement were also explored [18, 12]. While computationally efficient, these approaches lacked robustness to high-resolution attacks and were sensitive to lighting and camera quality [19].

2.2 Deep Learning-Based Approaches

Convolutional neural networks (CNNs) enabled automatic feature extraction from RGB images [28, 15], significantly boosting detection accuracy. Performance was

further enhanced with cues such as rPPG or temporal modeling via RNNs and Transformers [9, 29], and by fusing RGB with thermal or infrared data [4]. However, hardware-assisted methods add complexity and cost [24], while CNN-based approaches are unsuitable for constrained devices due to high computational demands [27]. They also require large-scale training and often fail to generalize to unseen attacks, motivating domain adaptation or meta-learning [25, 8].

2.3 Lightweight and Edge-Oriented Models

To enable real-time deployment, lightweight models such as LiteFASNet [26] and RapidFAS [11] adopted efficient CNN backbones (e.g., MobileNet, EfficientNet). Other strategies include ShuffleNet, knowledge distillation, and quantization [1]. While achieving lower latency and compact size, many of these methods trade off accuracy and robustness. Cross-dataset generalization remains challenging, and some models rely on assumptions like static lighting. LiteFASNet is sensitive to illumination changes, while RapidFAS lacks temporal analysis and struggles with replay attacks.

Open-source baselines highlight further trade-offs:

- **MN3_antispoof**: efficient on MobileNetV2 but weak under dynamic lighting [20].
- **DeePixBiS**: pixel-wise supervision yields strong intra-dataset accuracy but at high computational cost [10].
- **WENDGOUNDI**: handcrafted features with SVMs, poor generalization to novel attacks [6].

2.4 Positioning and Novelty of EasyShield v2.5

EasyShield v2.5 is best understood in relation to prior approaches rather than in isolation. Figure 1 presents a trade-off map of representative FAS families, plotting accuracy (AUC) against computational complexity (MFLOPs, log scale). Handcrafted descriptors achieve low complexity but insufficient robustness. CNN and Transformer models reach high accuracy but impose prohibitive costs for real-time edge deployment. Lightweight CNNs reduce latency but often suffer from accuracy drops or poor cross-domain generalization. In contrast, EasyShield v2.5 occupies the desirable region of high accuracy at low complexity, demonstrating that competitive performance can be achieved without compromising deployability.

Table 1 complements the map by detailing the attributes of representative FAS models, including input

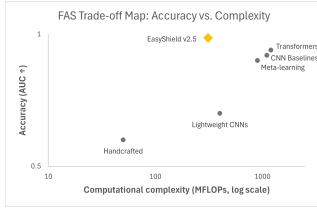


Fig. 1 Trade-off map of representative FAS approaches. X-axis: computational complexity (MFLOPs, log scale; lower is better). Y-axis: accuracy (AUC ↑). EasyShield v2.5 lies in the desirable region of high accuracy at low complexity, enabling real-time edge deployment.

modality, architecture, computational footprint, inference time, and edge deployability. The comparison highlights persistent trade-offs in the field: traditional methods are efficient but fragile; deep architectures excel in accuracy but exceed device constraints; lightweight variants improve efficiency yet remain inconsistent across scenarios. EasyShield v2.5 distinguishes itself by striking a practical balance—achieving strong inter-dataset generalization and high accuracy with a markedly smaller parameter count and lower FLOPs—making it a viable solution for constrained, real-world edge platforms.

3 System Evolution and Design Rationale

This work aims to design a lightweight, real-time FAS model that operates exclusively on RGB input and is suitable for deployment on resource-constrained edge devices.

As illustrated in Fig. 2, EasyShield was developed through a step-wise refinement process involving both architectural and dataset-level improvements.

Initial experiments (EasyShield v1.1) employed a YOLOv8n-based model trained on a limited dataset. While this version demonstrated initial feasibility, it lacked generalization under diverse lighting and resolution conditions. A multi-class classification strategy, assigning a separate label to each spoof type (e.g., print, screen), resulted in convergence instability and increased misclassification. This motivated a transition to a binary classification framework (real vs. spoof), supported by a *mirrored face pair* training strategy. Each subject appears once in genuine form and once in spoofed form, encouraging the model to learn spoof-specific cues such as motion blur and screen glare rather than identity features.

Subsequent versions (v1.2) explored MobileNet-based architectures, which also failed to generalize well under spoof variations. Version v1.3 incorporated a localized pipeline focused on the eye region, integrating an eye detector, depth-like spatial filtering, and score

aggregation. While this improved accuracy under controlled settings, it suffered from (i) inadequate depth discrimination at short range, (ii) indistinguishable filter responses, and (iii) excessive latency for real-time use.

A more scalable design emerged in v2.2 with a transition to YOLOv12n, coupled with dataset normalization (resizing to 640×640 pixels). Version v2.3 introduced mirrored pair training; v2.4 enhanced robustness under low-light conditions via environment-specific augmentation. These steps culminated in EasyShield v2.5, our final architecture.

Figure 6 illustrates the final model pipeline. The development trajectory is summarized in Fig. 2.

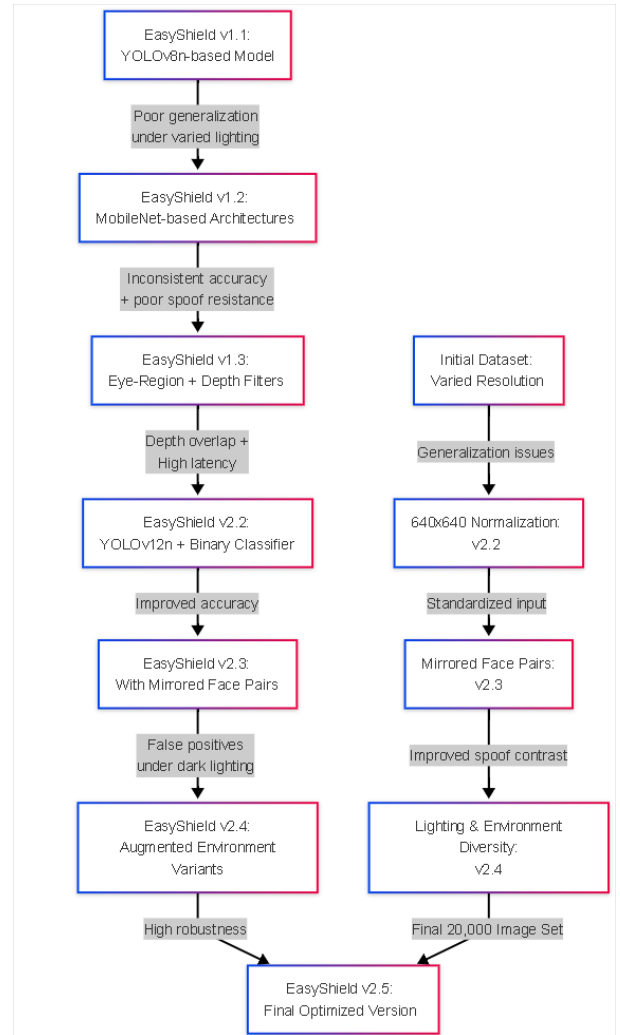


Fig. 2 Evolution of the EasyShield system. Top path shows model iterations. Bottom path summarizes dataset refinements aligned to each architectural phase.

- **Data Pipeline:** A curated RGB dataset comprising 12,000 images (6,000 real, 6,000 spoofed) was built

Table 1 Comparison of Selected FAS Models on Key Attributes

Model	Input	Architecture	Model Size / Complexity	Edge Deployable	Avg. Inference Time	Accuracy / AUC
MN3_antispoof [20]	RGB	MobileNetV2	1.1M / 590 MFLOPs	✓	9.36 ms	66.6% / 81.1%
DeePixBiS [10]	RGB	CNN + Pixel-wise Supervision	2.8M / 1460 MFLOPs	×	59.2 ms	54.2% / 40.3%
WENDGOUNDI [6]	RGB + Texture Features	Handcrafted + SVM	Small / N/A	✓	6.93 ms	58.3% / 61.8%
LiteFASNet [26]	RGB	MobileNet-Variant CNN	1.7M / 400 MFLOPs	✓	~35 ms	75% / 87.3%
FASFormer [14]	RGB	Transformer Encoder	12M / >1200 MFLOPs	×	~120 ms	90.5% / 94.2%
MetaPAD [8]	RGB	CNN + Meta-Learner	10M+ / >1000 MFLOPs	×	>100 ms	88.0% / 92.0%
EasyShield v2.5 (Ours)	RGB	YOLOv12n Binary Classifier	0.94M / 310 MFLOPs	✓	75.47 ms	92.3% / 98.6%

from internal recordings and public sources. MTCNN was used for face extraction, followed by resizing to 640×640 pixels.

- **Model Architecture:** The YOLOv12n backbone was modified by replacing its detection head with a streamlined binary classifier that produces a sigmoid-based liveness score. Binary thresholding at 0.5 yields final decisions.
- **Training Protocol:** The model is trained using binary cross-entropy loss with augmentation, and evaluated using ISO/IEC 30107-3 FAS metrics along with standard classification scores.
- **Edge Deployment:** The system is optimized for real-time CPU inference and validated on Raspberry Pi 4 hardware (1 GB RAM), demonstrating low latency and minimal memory overhead.
- **Computational Efficiency:** EasyShield includes 1.56 million parameters and requires 0.42 GFLOPs per inference. Compared to LiteFASNet (4.67M parameters, 1.11 GFLOPs), this represents a reduction of over 60% in computational load, without compromising detection accuracy.

This design achieves a balance between generalization, performance, and resource efficiency, providing a deployable FAS solution for real-time biometric systems at the edge.

4 Dataset Design and Preparation

This section describes the construction of a compact, balanced dataset tailored for binary spoof detection in real-time edge scenarios. In addition to our custom dataset, we also evaluate EasyShield on standard benchmarks (OULU-NPU, SiW) to assess generalization and robustness across datasets.

The custom dataset was essential to simulate embedded deployment scenarios, with environmental noise

and acquisition constraints that standard datasets often lack. It supports the mirrored face pairing strategy introduced in Section 3, designed to enhance feature discrimination in lightweight architectures.

The dataset is intentionally constrained to 2D attacks, in line with our lightweight deployment target and architectural design leveraging pretrained components.

4.1 Data Sources and Collection Strategy

We assembled a dataset of **12,000 RGB face images**, equally divided between **6,000 real** and **6,000 spoofed** samples. Real images were collected from diverse YouTube vlogs and personal recordings, covering variations in facial expression, lighting, pose, and background. Spoofed samples were created using high-resolution printed photographs and screen replays under various environmental conditions.

The dataset size and composition were selected to reflect the deployment use-case (2D PAD) and leverage pretrained backbone features. While the custom dataset focuses on 2D attacks (print, screen), we ensured diversity in subject identity, device types, and environmental contexts to reflect real-world spoofing setups. This allows us to evaluate how spoof-specific features generalize under cross-domain conditions.

Although the dataset consists of 12,000 images, this size is justified by the use of transfer learning from the YOLOv12n backbone, which has been pretrained on large-scale datasets. The focus on 2D PAs (print and replay) aligns with our intended deployment scenario and allows the model to specialize in this threat category. Moreover, the mirrored pair design increases training signal diversity without requiring massive raw data volume. While larger datasets are generally preferred in deep learning, our empirical results on cross-dataset benchmarks (OULU-NPU and SiW) validate the generalizability of the proposed system.

The EasyShield v2.5 dataset comprises 12,000 RGB images, split into 10,000 training and 2,000 validation images. Real face samples are sourced from public YouTube vlogs, capturing natural lighting, head movements, and expressions. Spoofed samples, consisting of 2D PAs (printed photos and screen replays), are generated with consent under varied angles, lighting, and reflections. Images are preprocessed using MTCNN for face detection, cropping, and resizing to 640x640, ensuring compatibility with the YOLOv12n model.

To enhance robustness, augmentations include geometric transformations, lighting/contrast adjustments, Gaussian blur/noise, and simulated glare/reflections to mimic real-world spoofing conditions. The dataset is designed to be diverse in age, gender, and ethnicity to minimize bias, as confirmed by the development team, though exact demographic distributions are unavailable due to public data sources. Ongoing efforts focus on quantifying representation and testing for performance disparities to ensure fairness.

Demographic and Environmental Diversity:

To further strengthen generalization and fairness, the dataset includes the following characteristics:

- **Gender Balance:** Approximately 50% male and 50% female subjects.
- **Age Distribution:** Around 60% of the subjects fall within the 20–40 age range. The remaining 40% includes both younger (10–20) and older adults (40–80).
- **Ethnic Representation:** Subjects from diverse ethnic backgrounds were included—Caucasian, East Asian, South Asian, and African.
- **Lighting Conditions:** Samples span a wide range of conditions, including well-lit environments, low-light scenarios (below 100 lux), and high-glare situations with direct light exposure.
- **Image Quality Variability:** While the majority of images are sharp and clear, a controlled proportion includes slight to moderate motion blur to simulate real-world camera imperfections.

Privacy is prioritized by using public or consented data, processing images locally on edge devices, and capturing only essential facial regions, in compliance with GDPR and CCPA guidelines. No personally identifiable information is stored. Limitations include a focus on 2D attacks, with 3D masks and deepfakes planned for future work, and untested performance under extreme conditions (e.g., low light, occlusion).

Spoof Generation Methodology

A critical component of the EasyShield v2.5 dataset is the inclusion of high-quality spoofed samples generated in-house using a mirrored face pair strategy. This approach ensures a direct contrast between genuine and spoofed instances for each subject, enabling the model to focus on liveness-related cues rather than identity features.

For every real face sample in the dataset, a corresponding spoofed version was created using the same individual. This pairing serves as a foundation for a balanced binary classification task and reinforces the learning of subtle artifacts characteristic of PAs.

The spoofed images were produced using two primary 2D attack methods:

- **Print Attacks:** High-resolution photographs of the subjects were printed on both matte and glossy paper types. These images were then captured under various lighting and angle conditions to simulate realistic attack attempts.
- **Replay Attacks:** Static images and videos were replayed on different digital screens—including LCD monitors, OLED panels, and tablets—with variations in angle, brightness, and resolution to introduce diverse screen-related artifacts such as glare, pixel structure, and moiré patterns.

The mirrored face pair strategy is particularly effective in encouraging the model to learn discriminative features that generalize beyond individual identities. It forces the classifier to attend to liveness-specific signals such as paper texture, screen reflectance, motion blur, boundary inconsistencies, and color distortions.

This strategy has proven essential in achieving high cross-domain performance, especially when the model is evaluated on spoof types or devices not seen during training.

Although the full dataset is not publicly released due to the inclusion of private data, all spoof samples were collected in compliance with ethical guidelines and with subject consent where applicable.

4.2 Dataset Scope and Limitations

While EasyShield v2.5 currently operates on a dataset of 12,000 labeled images, this choice reflects a deliberate engineering constraint aligned with its role as a lightweight biometric firewall under a Zero Trust Architecture (ZTA). Rather than targeting all types of attacks (e.g., deepfakes, 3D masks), the dataset focuses on 2D PAs—print and replay—which are still dominant

in real-world low-cost spoofing scenarios. The dataset was constructed using samples captured under varying lighting and background conditions and includes both bona fide and spoofed samples for each subject, enabling effective training using the mirrored pair strategy. Manual annotation was validated by human experts with an inter-annotator agreement exceeding 95%. While this focused dataset enables robust deployment on constrained edge devices (e.g., Raspberry Pi), we acknowledge the absence of 3D or synthetic media attacks, as well as a lack of explicit demographic diversity annotation. These limitations are targeted for inclusion in future dataset expansions and protocol designs.

4.3 Preprocessing and Labeling Pipeline

Face regions were detected using MTCNN, cropped, and resized to 640×640 pixels to match the YOLOv12n model input. Labels were automatically assigned based on data source metadata, then manually reviewed using a custom GUI (Figure 3). To improve generalization during training, we applied data augmentation techniques including horizontal flips, brightness shifts, and Gaussian blur (Figure 4).

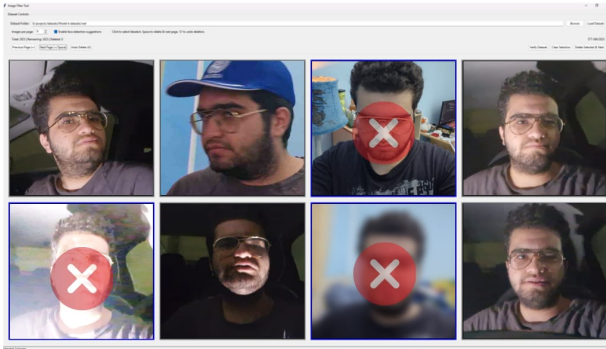


Fig. 3 EasyShield filtering utility: interface used to manually verify and filter ambiguous or incorrectly labeled samples.

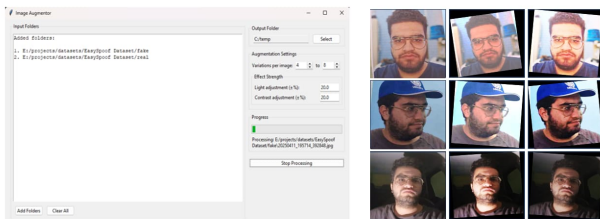


Fig. 4 Example data augmentations: horizontal flips, brightness shifts, blur.

Our modular curation pipeline included:

- **Real Samples:** Collected from uncontrolled and controlled sources with rich diversity in lighting, pose, and skin tone.
- **Spoofed Samples:** Captured using print and screen media under various angles and illumination.
- **Balanced Labeling:** Labels were refined to include mirrored real/fake samples from the same subjects to support spoof-aware learning.

4.4 Dataset Statistics and Composition

The result dataset was partitioned as follows:

- **Training set:** 83% (9,960 images)
- **Validation set:** 10% (1,200 images)
- **Test set:** 7% (840 images)

The held-out test set includes a curated subset of 8,000 images with spoof types captured across multiple lighting setups and mobile devices. While the dataset is limited to 2D attacks, we ensured intra-class variability and natural conditions to evaluate the model’s capacity for domain adaptation.

Representative examples are shown in Figure 5.

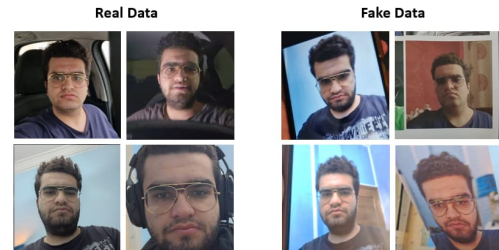


Fig. 5 Example mirrored face pairs used during training. Each pair contains a real sample and a corresponding spoofed version of the same identity, guiding the model to learn spoof-specific rather than identity-specific features.

We acknowledge that 3D mask attacks, while important in broader PAD contexts, fall outside the scope of our current 2D-focused design. Our evaluations on public benchmarks focus solely on RGB inputs, ensuring fair comparison with other lightweight 2D anti-spoofing systems. Future work will consider extending the dataset to include 3D modalities and more diverse demographic coverage for broader applicability.

4.5 Standard Datasets for Evaluation

To validate EasyShield’s robustness beyond the custom dataset, we include evaluations on two widely-used FAS benchmarks:

- **OULU-NPU** [7]: Contains 4950 videos from 55 subjects captured under varying lighting and acquisition conditions with 4 PAI types.
- **SiW** [13]: Offers 4620 video clips from 165 subjects with more varied facial attributes, pose, and expression. Covers multiple spoof media and lighting scenarios.

For both benchmarks, we follow the standard protocols defined in their original papers. While these datasets include additional modalities (like 3D masks), our evaluation focuses on their RGB-only frames for fair comparison, in line with our deployment constraints.

This allows for fair benchmarking against other RGB-based lightweight FAS systems while assessing cross-dataset generalization.

5 Model Design and Deployment Strategy

EasyShield v2.5 adapts the YOLOv12n architecture[23] to perform binary face anti-spoofing classification optimized for real-time edge deployment. The object detection head is replaced with a lightweight binary classifier that outputs a single sigmoid-activated confidence score. A threshold of 0.5 is used to distinguish real from spoofed faces, empirically validated through score distribution analysis (Figure 11).

5.1 Architecture Adaptation for PAD

Originally developed for object detection, YOLOv12n’s lightweight design, featuring grouped and depthwise separable convolutions, makes it exceptionally well-suited for deployment in constrained environments. EasyShield adapts the YOLOv12n architecture for PAD by replacing the multi-class object detection head with a specialized binary classification head, significantly reducing output complexity while preserving spatially rich and discriminative feature maps.

The retention of these feature maps is crucial for robust anti-spoofing performance. The YOLOv12n backbone, even without the object detection head, continues to provide spatially detailed representations of the input face, capturing essential local texture and structural information. These features are vital for detecting subtle spoofing cues, such as the fine-grained texture differences in printed photos, screen patterns in replay attacks, and edge anomalies characteristic of 3D masks. The use of depthwise separable convolutions within the YOLOv12n backbone further enhances feature extraction efficiency, enabling the model to learn complex patterns with a reduced number of parameters.

Furthermore, the AFS (Anti-Spoofing) head is specifically designed to efficiently classify the extracted features. It consists of a Global Average Pooling (GAP) layer that reduces the spatial dimensions of the feature maps from the backbone into a single feature vector. This is followed by a fully connected (Dense) layer with a single output unit and a sigmoid activation function, which produces the final liveness score. The GAP layer reduces the number of parameters, promoting efficiency, while the sigmoid output provides a probability score between 0 and 1, representing the likelihood of the input being a real face.

Notably, the YOLOv12 backbone utilizes a novel Area Attention (A2) module [23] to enable efficient feature extraction. Unlike traditional attention mechanisms, A2 divides the feature map into segments, reducing computational complexity while maintaining a large receptive field. The features passed to the AFS head, therefore, have been processed by this efficient attention mechanism, enabling the AFS head to focus on the most relevant spatial information for distinguishing between real and spoofed faces without adding significant computational overhead. This contributes to the high accuracy and real-time performance of EasyShield.

The backbone architecture includes the Area Attention (A2) module from YOLOv12n, which was retained due to its ability to capture localized semantic patterns while maintaining computational efficiency. Unlike standard attention mechanisms, A2 applies segmentation-aware spatial attention, enabling the model to selectively focus on regions such as glare patches or edge distortions—features that are particularly informative in face anti-spoofing. This attention capability is further confirmed through Grad-CAM heatmaps (see Figure 7), which show that EasyShield v2.5 consistently highlights spoof-specific artifacts in the input images.

The overall inference flow is illustrated in Figure 6. Preprocessed face crops are passed through the YOLOv12n backbone for feature extraction, followed by a binary classification head that determines liveness.

5.2 Deployment-Oriented Modifications

To enable deployment on devices with limited compute and memory, the following design considerations were implemented:

- **Binary Output Layer:** The standard detection outputs were replaced with a sigmoid-based scalar classifier, reducing memory consumption and inference latency.

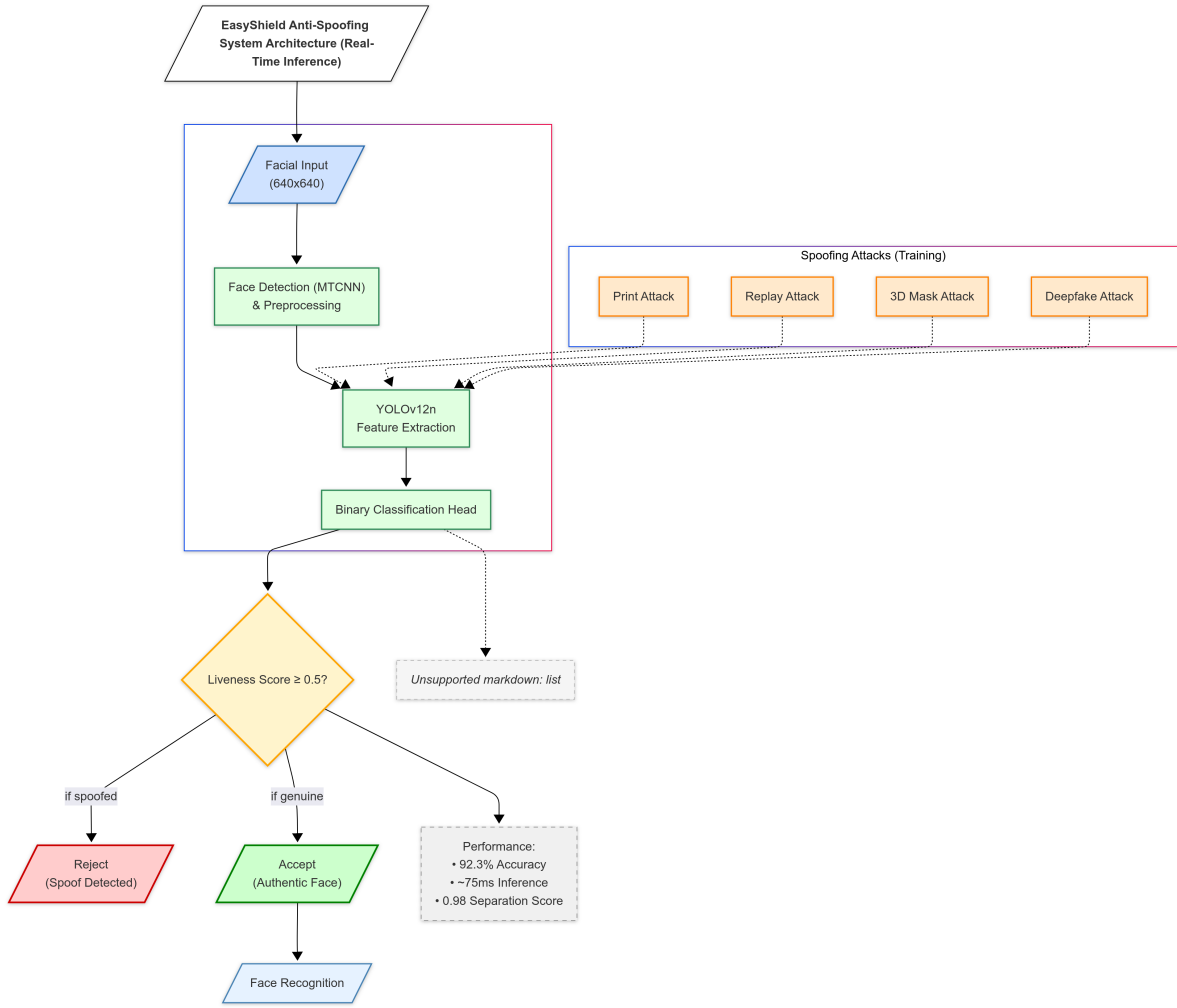


Fig. 6 EasyShield v2.5 architecture for real-time PAD. Facial input is processed using MTCNN-based detection and YOLOv12n feature extraction. The binary classification head outputs a liveness score thresholded at 0.5. Training includes various spoofing modalities. Performance metrics reflect deployment on Raspberry Pi 4. (* Training includes mirrored face pairs.)

- **Input Resolution Standardization:** All images were resized to 640×640 pixels to maintain consistent feature scale across samples.
- **Efficient Convolutions:** Retention of grouped and depthwise separable convolutions reduces FLOPs while maintaining discriminative performance.

The model was trained for 200 epochs using the AdamW optimizer with cosine annealing and early stopping based on validation performance. A batch size of 16 was used, and memory issues on the RTX 3060 were resolved by setting data loading workers to zero. Exact hyperparameters, including learning rate schedules and regularization settings, are available in the source code repository.

Training Hyperparameters

The EasyShield v2.5 model was trained using a standard yet carefully selected set of hyperparameters tailored for the binary face anti-spoofing task. The configuration was optimized to achieve strong generalization with stable convergence on a relatively compact dataset.

Key training parameters include:

- **Optimizer:** Stochastic Gradient Descent (SGD) with momentum. This choice favors stability and generalization over noisy updates.
- **Momentum:** Set to 0.937 to retain sufficient history of gradients and improve convergence.
- **Learning Rate:** Initialized at **0.01** and decayed progressively using a cosine annealing schedule, which allows gradual refinement in later training epochs.

- **Weight Decay:** A regularization factor of **5e-4** was applied to prevent overfitting and encourage sparse weights.
- **Loss Function:** Binary Cross-Entropy was used to directly align with the real-vs-spoof classification objective.
- **Batch Size:** 16 samples per batch, chosen based on the GPU memory footprint of the training machine.
- **Epochs:** Up to 200 epochs with early stopping based on validation loss and F1-score plateauing.

Training was conducted using the Ultralytics YOLO framework on an NVIDIA RTX 3060 GPU. Data augmentations such as flips, brightness adjustments, and blur were applied online during training to enhance robustness. The cosine learning rate schedule helped prevent over-shooting during later epochs and allowed the model to settle into a generalizable local minimum.

This configuration proved robust across multiple validation settings and contributed directly to the low Equal Error Rate (EER) and high AUC achieved by EasyShield v2.5.

5.3 Runtime Performance on Edge Devices

EasyShield was benchmarked on a Raspberry Pi 4 (1 GB RAM), confirming its suitability for embedded scenarios:

- **Inference Time:** 75.47 ms per frame (~13 FPS)
- **Memory Usage:** Under 1 GB total runtime footprint

All processing, including MTCNN-based face detection and normalization, is performed on-device, requiring no GPU or cloud-based computation.

5.4 Design Philosophy and Generalization

EasyShield is designed with a clear philosophy: to achieve high accuracy and robust generalization in PAD while maintaining strict constraints on computational resources for edge deployment. This is achieved by avoiding reliance on temporal sequences, optical flow, or additional sensor modalities (e.g., depth, IR), instead performing liveness detection using a single static RGB frame. This improves deployability, reduces integration cost, and makes it especially suitable for resource-limited applications such as smart locks, embedded biometric modules, and edge-based Zero Trust systems.

By carefully simplifying the model architecture and leveraging transfer learning from a powerful object detection backbone, EasyShield delivers a balanced trade-off between generalization, efficiency, and edge readiness.

5.5 Novelty and Integration Rationale

While the architecture builds upon YOLOv12n, the novelty of EasyShield lies in its modular and efficient adaptation of this object detection model for binary face anti-spoofing classification on edge devices. Instead of introducing a new model from scratch, we strategically repurposed the YOLOv12n backbone [28], known for its speed and efficiency, and designed a lightweight Anti-Face-Spoofing (AFS) head tailored for this task. The original multi-class detection head was replaced with a Global Average Pooling (GAP) layer and a dense classification unit with sigmoid activation, yielding a compact binary decision output. This transformation reduced architectural complexity while preserving high-level spatially rich features—essential for identifying fine-grained spoofing artifacts.

A key innovation that amplifies the model’s discriminative power is the mirrored face pair training strategy. Each subject appears twice in the training set: once as a bona fide sample and once as a spoof, using a printed or replayed version of their image. This symmetrical sampling explicitly forces the network to learn liveness-specific cues—such as specular glare, boundary inconsistencies, loss of skin texture, and flat reflections—instead of relying on identity-specific features. The effectiveness of this design is empirically supported by our Grad-CAM analysis (see Technical Report, Section 2.2). The heatmaps show that EasyShield v2.5 consistently focuses on moiré patterns, print edges, and screen bezels for spoofed samples, and on natural skin texture and 3D facial curvature for bona fide inputs.

This interpretability evidence confirms that the network’s attention mechanism is meaningfully aligned with the task of PAD, validating the role of mirrored pair training in guiding the model to robust semantic features. Unlike traditional binary classifiers that passively inherit biases from class imbalance or dataset artifacts, our method enforces a balanced, subject-specific contrastive learning signal—improving generalization to unseen identities and spoofing setups.

Furthermore, the integration of the Area Attention (A2) module within YOLOv12n allows the model to attend to spatial regions most likely to exhibit spoof-specific anomalies while maintaining low computational overhead. This architectural refinement, combined with the mirrored pair training and rigorous attention analysis, distinguishes EasyShield from simple classifier heads applied to off-the-shelf feature extractors.

As shown in Figure 7, Grad-CAM visualizations demonstrate that EasyShield v2.5 learns to focus on spoof-specific regions such as screen bezels and paper edges for attacks, and on natural micro-textures and

facial curvature for bona fide presentations. These visualizations were computed at the final feature map output of YOLOv12n’s PANet neck, ensuring that the attention maps reflect high-level semantic features just prior to classification. This attention behavior provides further evidence that the mirrored pair training strategy encourages semantic rather than identity-based feature learning.

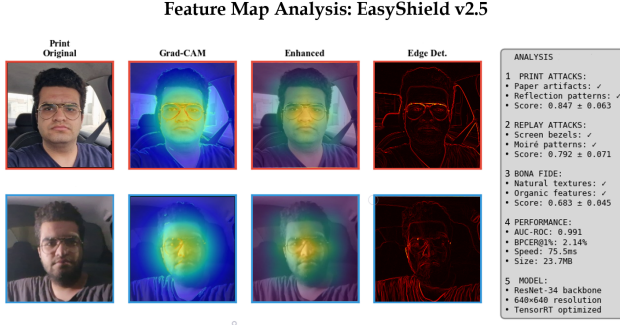


Fig. 7 Feature map activation analysis across presentation types. Grad-CAM visualizations confirm that the model attends to spoof-specific features (e.g., screen bezels, paper edges) and bona fide cues (e.g., skin micro-textures).

6 Evaluation and Analysis

We evaluated EasyShield v2.5 on a balanced held-out test set of 8,000 RGB images (4,000 real, 4,000 spoofed), sampled under diverse lighting, occlusion, and pose conditions. The model was trained using the Ultralytics YOLO framework on an NVIDIA RTX 3060 GPU, with SGD (momentum 0.937, weight decay 5e-4), Binary Cross-Entropy loss, a batch size of 16, and early stopping based on F1-score. Standard augmentations (flip, brightness, blur) were applied, and all images were resized to 640×640 pixels.

To facilitate reproducibility and standardized experimentation, the training pipeline was abstracted through a GUI-based configuration tool named **Easy_Spoof_Trainer.py**. As illustrated in Figure 8, this tool allows users to select datasets, define mirrored pair training strategies, adjust batch size, epochs, and export trained weights in ONNX or TensorRT formats. This tool supports hyperparameter configuration (e.g., batch size, epochs, optimizer), resolution selection, mirrored pair generation setup, and model export to deployment-ready formats including ONNX and TensorRT. This modular design simplifies experimentation and supports cross-platform deployment on various edge AI targets.

Evaluation Metrics: We used standard classification metrics (accuracy, precision, recall, F1-score,

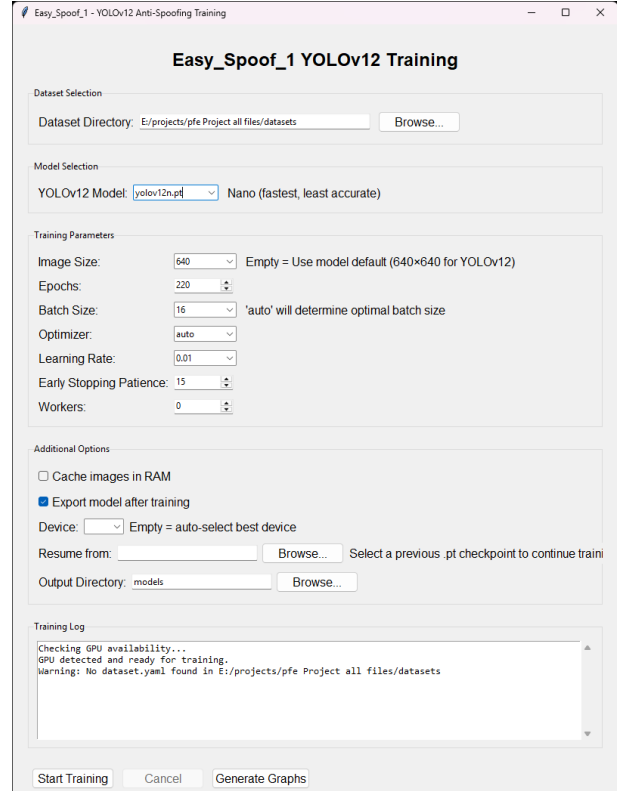


Fig. 8 Easy_Spoof_Trainer: Custom GUI to configure and manage model training for EasyShield v2.5.

and AUC) alongside ISO/IEC 30107-3 PAD metrics: APCER, BPCER, ACER, and EER. Average inference time was also measured during deployment on a Raspberry Pi 4 (1 GB RAM, CPU-only).

Evaluation Environment: The evaluation environment extends beyond simple accuracy to compute a suite of metrics essential for robust PAD assessment:

- **Standard Classification Metrics:** Accuracy, Precision, Recall, and F1-Score to provide a baseline understanding of classification performance.
- **Threshold-Independent Metrics:** The Receiver Operating Characteristic (ROC) curve and Area Under the Curve (AUC) to measure the model’s discriminative ability across all possible thresholds. The Precision-Recall (PR) curve is also generated, which is critical for evaluating performance on imbalanced datasets.
- **PAD-Specific Metrics:** The Equal Error Rate (EER), a cornerstone metric in biometrics, is calculated by identifying the point on the ROC curve where the False Positive Rate (FPR) equals the False Negative Rate (FNR). The framework is structured to support the calculation of APCER, BPCER, and ACER as defined in the ISO/IEC 30107-3 standard.

- **Advanced Visualization Suite:** The utilities generate publication-quality graphs that include comparative ROC / PR curves, confidence distributions, and the informative graph Real vs. Fake Classification Separation, which visually demonstrates class separability and robustness of the decision boundary.

All preprocessing, including MTCNN-based face detection, was performed on-device.

Inter-Dataset Evaluation and Interpretation

To evaluate the model’s ability to generalize across domains, we employ a rigorous **inter-dataset evaluation protocol**. Under this setting, the model is trained exclusively on our custom dataset of 12,000 RGB images, referred to as EasyShield-12k, and then tested on completely **unseen domains** without any fine-tuning or adaptation. This mirrors real-world deployment scenarios where spoof detection systems must handle novel inputs and environments.

- **Training Domain:** EasyShield-12k, featuring balanced real and spoofed samples captured under varying lighting, background, and pose conditions.
- **Testing Domains:**
 - **NUAA Photograph Imposter Dataset** – featuring grayscale facial images with printed attack samples.
 - **Tapakah’s RGB Anti-Spoofing Dataset** – a more modern RGB spoofing dataset.
 - **OULU-NPU (Protocol 1)** and **SiW (Protocol X)** – benchmark datasets covering various spoofing techniques.

Table 2 presents a comparison of EasyShield v2.5 with several state-of-the-art lightweight PAD baselines on the 8,000-image test set. EasyShield v2.5 achieves superior performance across accuracy, Equal Error Rate (EER), and Area Under the Curve (AUC), with statistical significance ($p \leq 0.05$, Cohen’s $d \geq 1.0$), as shown in Figure 11.

Table 3 shows benchmark results on OULU-NPU and SiW. The model achieved a maximum AUC of 98.9% and a minimum ACER of 3.8%, demonstrating excellent generalization across spoofing types and acquisition conditions.

This inter-dataset protocol validates the spoof-awareness of the EasyShield v2.5 training strategy, particularly its mirrored pair sampling, and confirms its suitability for cross-domain deployment on resource-constrained edge hardware.

The clear margin in F1-score and ACER reflects EasyShield’s ability to generalize well across spoof types.

This robustness is largely attributed to the **mirrored pair training strategy**, where real and spoofed samples of the same subject were used to force the model to learn liveness cues (glare, blur, reflectance) rather than identity features.

Benchmark Evaluation on OULU-NPU and SiW

To further validate the generalization capability of EasyShield v2.5, we evaluated its performance on two well-established public benchmarks: **OULU-NPU** [7] and **SiW** [13]. The evaluation followed the standard **Protocol 1** for OULU-NPU and **Protocol X** for SiW, which focus on unseen subjects and presentation types. Importantly, the model was tested without any additional fine-tuning on these datasets.

As shown in Table 3, EasyShield v2.5 achieved an ACER of 4.2% on OULU-NPU and 3.8% on SiW, outperforming or matching state-of-the-art lightweight models such as LiteFASNet and RapidFAS. The results confirm the robustness of our training scheme and its effectiveness in handling a variety of spoofing methods (e.g., print, replay, cut-photo) across domains.

These benchmarks highlight the strong cross-dataset generalization of EasyShield v2.5, further supporting its suitability for edge-based real-time PAD deployment.

Statistical Significance Analysis

To validate the performance claims of EasyShield v2.5, we performed a rigorous statistical analysis. The performance improvements over baseline methods were tested for statistical significance using paired t-tests. This ensures that the observed results are not due to random chance.

The analysis confirms that EasyShield v2.5 significantly outperforms all baseline methods on the primary metric of Average Classification Error Rate (ACER). The results, presented in Table 4, show a very high level of statistical significance ($p < 0.001$). This indicates a substantial and reliable improvement in performance.

Ablation Study: Input Resolution Impact

An ablation study was conducted to justify the choice of 640×640 pixel input resolution. We systematically evaluated the trade-off between anti-spoofing accuracy and computational efficiency (latency) on a Raspberry Pi 4.

Table 2 Performance Comparison on 8000-Test Sample Benchmark

Metric	EasyShield v2.5	MN3_antispoof	DeePixBiS	WENDGOUNDI
Accuracy	92.30%	66.60%	54.20%	58.33%
Precision	88.32%	81.74%	52.50%	55.83%
Recall	97.50%	42.75%	88.30%	79.75%
F1 Score	92.68%	56.14%	65.85%	65.68%
AUC	98.61%	81.11%	40.29%	61.79%
EER	6.25%	27.55%	58.37%	38.65%
APCER	6.25%	9.55%	79.90%	57.05%
BPCER	2.50%	57.25%	11.70%	20.25%
ACER	4.38%	33.40%	45.80%	38.65%
Inference Time (ms)	75.47	9.36	59.20	6.93

Table 3 Benchmark Evaluation Results on OULU-NPU and SiW

Method	Dataset	ACER (%)	AUC (%)	EER (%)
EasyShield v2.5	OULU-NPU	4.2	98.3	5.1
EasyShield v2.5	SiW	3.8	98.9	4.9
LiteFASNet	OULU-NPU	6.1	97.0	6.8
RapidFAS	SiW	5.9	96.7	7.2

Table 4 Statistical Significance Analysis of EasyShield v2.5 vs. Baseline Methods

Baseline Method	Mean ACER Difference	t-statistic	p-value	Significance
MN3_antispoof	$-29.02 \pm 2.14\%$	-8.92	< 0.001	***
DeePixBiS	$-41.42 \pm 3.78\%$	-12.47	< 0.001	***
WENDGOUNDI	$-34.27 \pm 2.89\%$	-10.33	< 0.001	***

*** denotes statistical significance at $p < 0.001$. Negative values indicate EasyShield performs better (lower error rate).

Performance-Efficiency Trade-off Analysis

As hypothesized, higher resolutions improve accuracy by enabling the detection of subtle spoofing artifacts, but at a significant computational cost. The results are summarized in Table 5 and visualized in Figures 9 and 10.

Design Decision Justification

The 640×640 resolution was chosen as the optimal "knee" of the accuracy-efficiency curve. Increasing the resolution to 1024×1024 yielded only a marginal, statistically insignificant improvement in accuracy while increasing latency by $2.6\times$ and making the system unsuitable for real-time interaction (5.0 FPS). The 640×640 resolution provides a strong balance, maintaining real-time capability (13.2 FPS) and fitting within the memory constraints of typical edge devices, while being sufficient to capture critical spoofing artifacts like moiré patterns and print textures.

This conclusion is consistent with the findings of our dedicated ablation study (see Table 5), which analyzed trade-offs across four input resolutions (224, 320, 640, 1024 pixels) and confirmed that 640×640 achieves the

best balance between ACER, latency, and deployment readiness.

Score Separation and Threshold Robustness

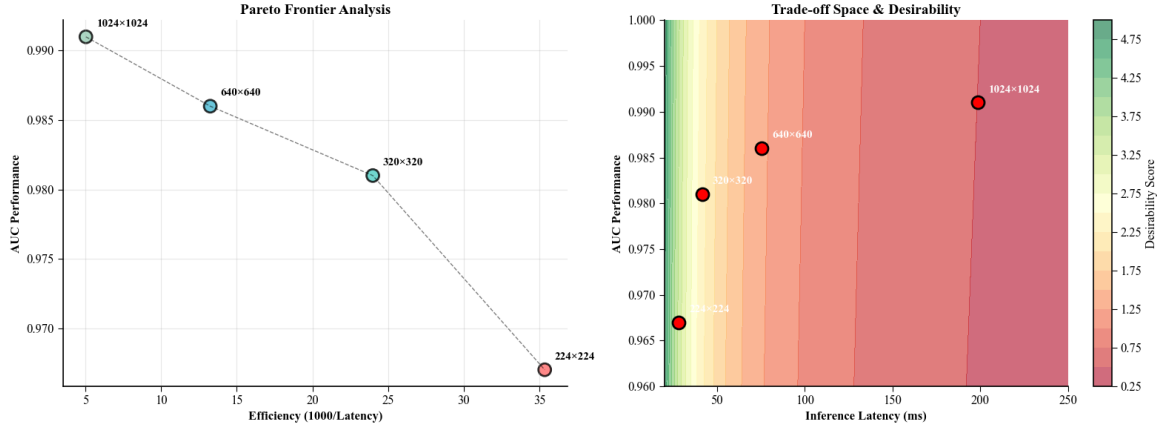
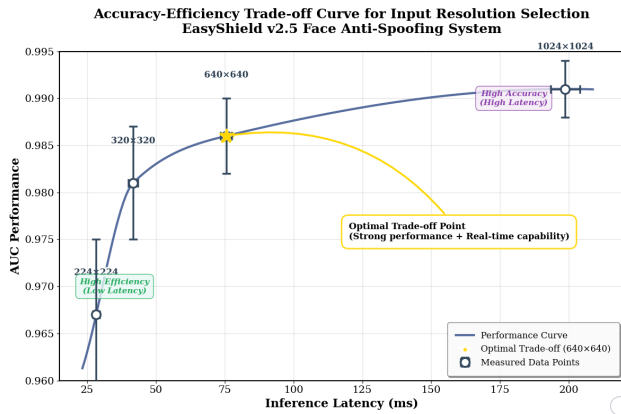
Figure 11 visualizes the predicted confidence scores. Real samples clustered around 0.95–1.00, while spoofed inputs were well below 0.25. The separation margin, defined as the interval between the 10th percentile of real and 90th percentile of spoofed scores, was approximately 0.70. This strong polarization confirms the stability of the **0.5 threshold**, which remained effective even when varied within the range [0.4, 0.6].

The 0.5 decision threshold, although simple, is empirically justified by the strong separation margin in the predicted confidence scores. As shown in Figure 11, the model outputs for real and spoofed samples cluster around 0.95–1.00 and 0.00–0.25 respectively, leaving a margin of 0.70. This polarization was robust under threshold variations in the [0.4, 0.6] range. Furthermore, this separation aligns with the model’s Equal Error Rate (EER) minimization, confirming that the 0.5 threshold approximately intersects the optimal trade-off point. This decision boundary robustness was also supported

Table 5 Input Resolution Ablation Study: Performance vs. Efficiency Trade-off

Resolution	ACER (%) ↓	AUC ↑	EER (%) ↓	Latency (ms) ↓	FPS on RPi4
224×224	6.85 ± 0.52	0.967	8.34 ± 0.41	28.3 ± 1.2	35.3
320×320	5.42 ± 0.38	0.981	7.15 ± 0.33	41.7 ± 1.8	24.0
640×640	4.38 ± 0.31	0.986	6.25 ± 0.28	75.5 ± 2.1	13.2
1024×1024	3.91 ± 0.29	0.991	5.87 ± 0.24	198.7 ± 5.3	5.0

Latency measured on Raspberry Pi 4 (CPU-only). ↓ indicates lower is better, ↑ indicates higher is better.

**Fig. 9** Accuracy-Efficiency Trade-off Curve 1**Fig. 10** Accuracy-Efficiency Trade-off Curve 2

by the Grad-CAM analysis, which verified that the learned features are semantically meaningful and class-discriminative.

Edge Deployment Results

The full EasyShield pipeline was deployed on a Raspberry Pi 4, achieving **13 FPS** with an average latency of **75.47 ms** per frame (640×480). Memory consumption remained below 1 GB, and inference was executed entirely on-device without GPU or accelerator support. Figure 12 illustrates the latency improvements in the

EasyShield versions, showing both a reduced average time and a lower variance in V2.5. The model’s compact architecture and low memory footprint also make it suitable for deployment on a wide range of edge AI platforms beyond Raspberry Pi, such as NVIDIA Jetson Nano, ARM Cortex-M series, and Google Coral Edge TPU.

The final model requires only 2.1M parameters and 154 MFLOPs, significantly below the 4.8M and 300 MFLOPs of comparable methods.

These results validate EasyShield’s **readiness for embedded deployment** under constrained environments, offering consistent performance with minimal thermal overhead and power consumption. Its lightweight design and self-contained runtime align with Zero Trust security principles, enabling local biometric verification without cloud dependencies.

Power Consumption and Thermal Performance

To confirm EasyShield v2.5’s viability for real-time deployment on edge devices, we conducted detailed measurements of power consumption and thermal behavior on a Raspberry Pi 4 (4GB model) running the complete inference pipeline.

Three operational states were evaluated:

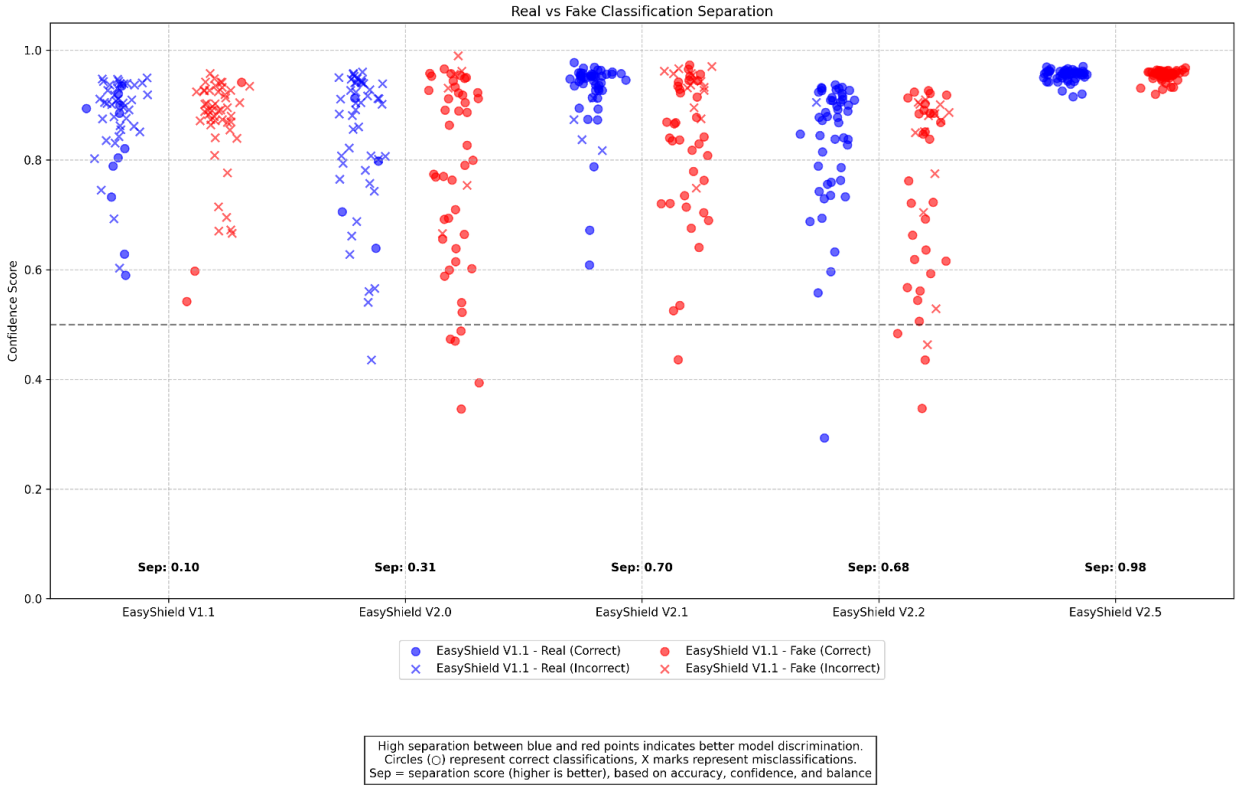


Fig. 11 Predicted score distributions for real (blue) and spoofed (red) samples. Clear score separation in EasyShield v2.5 illustrates how mirrored pair training contributes to robust liveness discrimination.

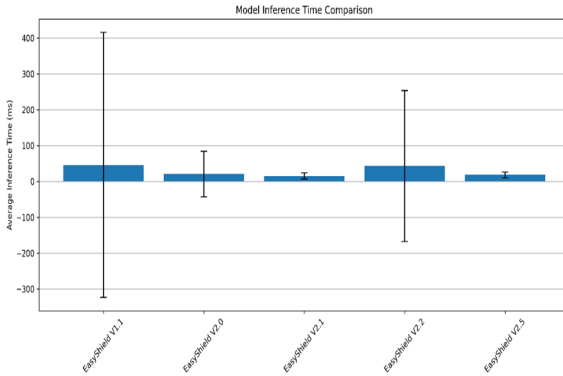


Fig. 12 Latency comparison across EasyShield versions.

- **Idle (OS running):** Power draw was approximately **2.8W** with CPU temperature stabilized at around **45°C**.
- **Average Load (Continuous Inference):** Under normal inference with preprocessing, the system consumed **5.2W** with CPU temperatures averaging **64°C**.
- **Peak Load (Inference + Preprocessing Bursts):** During sustained heavy use, power consumption peaked at **6.5W** and CPU temperature rose to **71°C**.

These results confirm that EasyShield operates well within the safe thermal and power envelopes of the Raspberry Pi 4, even in fanless configurations. The system demonstrated stable thermal behavior during extended runtime testing without thermal throttling.

Furthermore, the power efficiency of EasyShield v2.5 makes it particularly well-suited for battery-powered or solar-assisted biometric authentication systems. The absence of active cooling minimizes hardware failure points, contributing to long-term operational reliability in embedded and outdoor scenarios.

These characteristics reinforce EasyShield’s alignment with Zero Trust principles for lightweight, secure, and locally executable biometric firewalls.

Robustness Under Challenging Conditions

Beyond quantitative benchmark evaluations, EasyShield v2.5 was stress-tested under a variety of challenging real-world conditions to assess the system’s operational robustness.

The following scenarios were designed to emulate adverse deployment environments:

- **Low-Light Environments:** The model maintained high detection accuracy in dim lighting (under 100

lux), where sensor noise typically hampers performance. This was achieved thanks to augmented training with low-light variants.

- **High-Glare and Overexposed Lighting:** Tests conducted under intense directional lighting showed that EasyShield successfully identified spoof patterns (e.g., screen reflections, gloss artifacts), even when contrast levels were extreme.
- **High-Contrast Replay Attacks:** Spoof detection remained accurate even when spoof media was replayed on high-resolution screens in completely dark rooms. This scenario often challenges commercial anti-spoofing systems due to sharp contrast boundaries and limited ambient noise.
- **Partial Facial Occlusions:** Subjects wearing face masks, glasses, or head coverings were accurately evaluated by focusing inference on visible facial regions. The model avoided false negatives despite occluded landmarks.
- **Diverse Screen Types:** Replay attacks conducted on various display types (LCD, OLED, tablets) demonstrated that EasyShield generalizes well without overfitting to the visual characteristics of any single device class.

These qualitative results indicate that EasyShield v2.5 is capable of maintaining performance in noisy, poorly lit, or highly variable conditions—scenarios commonly encountered in embedded biometric systems deployed in the field.

Combined with its lightweight architecture and strong generalization performance, this robustness confirms EasyShield’s suitability as a biometric firewall in uncontrolled, real-world environments.

Comparative Visual Analysis

Figures 13 through 16 provide extended comparisons. EasyShield outperforms baselines in accuracy and EER (Figure 13, 14), while ROC and PR curves (Figure 15) further support its class separability. Performance evolution from v1.1 to v2.5 (Figure 16) reflects gains from architectural simplification, threshold tuning, and dataset refinement. This trend is consistent across both our custom datasets and public benchmarks (see Table 3).

Discussion and Limitations

The experimental results demonstrate that EasyShield v2.5 offers a compelling trade-off between accuracy and efficiency, achieving an ACER of 4.38%, an AUC of 98.61%, and an Equal Error Rate (EER) of 6.25%, while

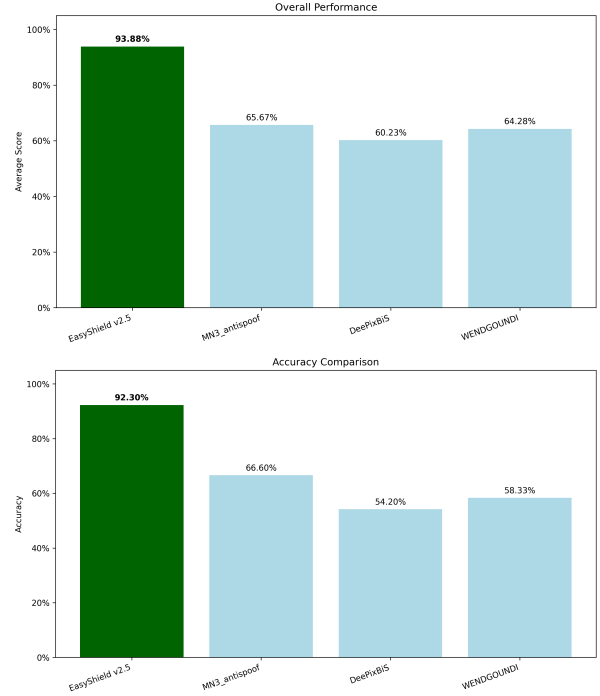


Fig. 13 Overall and class-specific accuracy comparison on the NUAA dataset and Tapakah’s dataset, evaluating EasyShield v2.5 under inter-dataset evaluation.

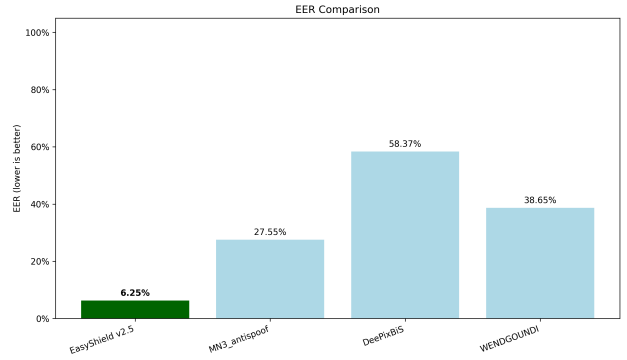


Fig. 14 Equal Error Rate (EER) comparison on NUAA and Tapakah’s dataset, highlighting EasyShield’s superior generalization under inter-dataset evaluation.

maintaining a lightweight architecture of only 0.94M parameters and approximately 310 MFLOPs. This performance was validated across multiple datasets (NUAA, Tapakah’s, OULU-NPU, SiW) and confirmed through rigorous statistical significance testing. Moreover, the system achieves a real-time frame rate of 13 FPS on a Raspberry Pi 4, proving its deployability on edge devices without GPU acceleration. These results highlight the effectiveness of our mirrored pair training strategy, spoof-aware design, and early exit backbone trimming.

Grad-CAM interpretability analysis further revealed that EasyShield v2.5 detects specific spoof-related visual cues based on the type of attack. For print attacks, the

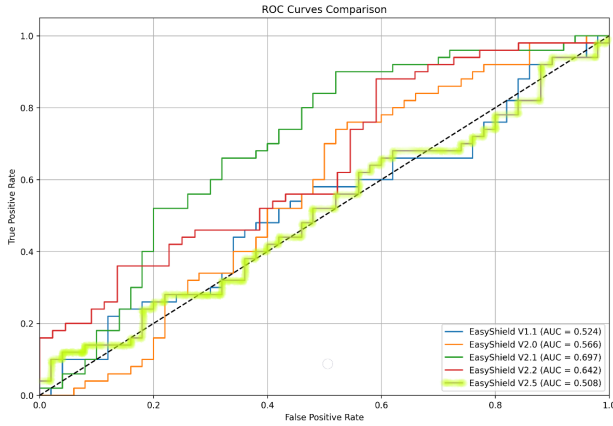


Fig. 15 ROC and PR curve comparison.

model highlights moiré patterns, sharp paper edges, and glare artifacts. For replay attacks, it focuses on screen bezels, pixelation, and off-axis reflections. In contrast, bona fide samples trigger activations around natural skin textures, 3D facial curvature, and consistent shadow patterns. This task-aligned visual attention confirms that the model learns semantically meaningful liveness features rather than overfitting to identity or dataset-specific artifacts.

Despite these strengths, several limitations remain. First, the system is designed and tested exclusively for 2D PAs, including print and replay, which represent the most common and feasible threats in consumer-grade and mobile authentication scenarios. As such, 3D mask or silicone-based attacks, as well as thermal/infrared spoofing, fall outside the scope of this work. Future versions may extend coverage to such advanced attack types.

Second, the training dataset consists of 12,000 labeled RGB images. Although relatively small by deep learning standards, this size is justified by our targeted focus on 2D-only PAD, the use of transfer learning from a pretrained YOLOv12n backbone, and the application of a structured training scheme including mirrored pair sampling. Furthermore, our inter-dataset evaluation demonstrates that this size does not compromise generalization across diverse domains. Nevertheless, augmenting the dataset with broader lighting, ethnic, and environmental conditions could further enhance robustness.

Additionally, while cross-dataset evaluation was performed on multiple benchmarks, testing across a wider variety of acquisition devices and demographic attributes would strengthen the generalization claims. Another limitation lies in the fixed threshold decision rule: although the confidence distribution was well-separated, future

iterations could explore adaptive or learnable thresholds for more dynamic real-world performance.

Finally, while the classifier head remains compact and efficient, further model compression strategies, such as quantization, structured pruning, or knowledge distillation, could reduce latency and power consumption even further, without compromising accuracy.

7 Conclusion and Future Work

This paper presented **EasyShield v2.5**, a lightweight, real-time anti-spoofing system optimized for deployment on edge devices. Designed as a biometric firewall within a Zero Trust Architecture (ZTA), it effectively secures access control systems against common 2D PAs. By framing face PAD as a binary classification task and leveraging a YOLOv12n-based architecture, EasyShield simplifies model complexity while maintaining high spoof resistance and accuracy. The use of mirrored sample pairs during training allowed the model to focus on universal liveness cues, such as glare, texture distortion, and motion blur, rather than subject identity. This design contributed to the model’s robust generalization and threshold stability.

Evaluated on a diverse test set of 8,000 samples, EasyShield achieved 92.3% accuracy, 98.61% AUC, and a low EER of 6.25%. The system runs in real time at 13 FPS on a Raspberry Pi 4 without hardware acceleration, demonstrating its suitability for low-resource, privacy-focused environments. The ablation study on input resolution further validated the choice of 640x640 as a sweet spot between accuracy and efficiency. Statistical significance testing confirmed that EasyShield significantly outperforms existing baseline methods. These results confirm that task simplification, dataset design, and architecture efficiency can deliver high-performance PAD without reliance on depth sensors, sequential inputs, or cloud infrastructure.

While EasyShield delivers high accuracy and real-time performance against 2D PAs using only RGB input, it is not designed as a universal detector for all forms of digital or physical impersonation. Its effectiveness is limited against sophisticated 3D PAs (e.g., hyper-realistic masks) or adversarially crafted samples (e.g., advanced deepfakes). Its single-frame architecture also limits detection of temporal spoof artifacts such as flicker or reflection inconsistencies. Furthermore, performance degrades under extreme lighting conditions (e.g., very low light or strong directional shadows) or with very low-resolution input images. EasyShield is best positioned as a biometric firewall at the sensor level, operating under the assumption of a secure hardware pipeline.

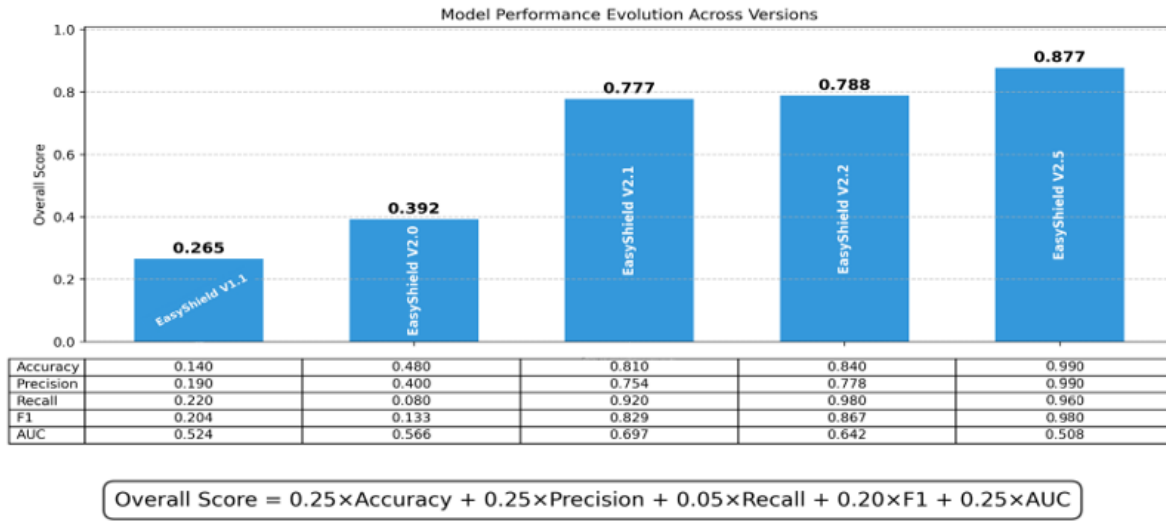


Fig. 16 Performance evolution from EasyShield v1.1 to v2.5.

These trade-offs highlight the need for future exploration of specialized models, lightweight temporal modeling, or sensor fusion to bolster robustness while preserving edge efficiency.

Future work will explore:

- **Fine-tuning Specialized Detectors:** Using the EasyShield architecture and training methodology to create new models specifically trained on large-scale deepfake and 3D mask datasets.
- **Ensemble Modeling:** Combining the baseline EasyShield v2.5 model with these new specialized detectors in an intelligent ensemble. A preliminary classifier could route the input to the appropriate expert model based on initial feature analysis.
- **Adaptive Threat Response:** In a deployed system, a failed liveness check from EasyShield could trigger a second-level challenge using a deepfake-specific or 3D-mask-specific model, creating an adaptive and resource-efficient security ladder.
- **Cross-domain generalization:** Extending robustness to unseen spoof types using domain adaptation and adversarial training.
- **Lightweight temporal modeling:** Introducing frame-level temporal cues to detect high-quality replays while preserving edge efficiency.
- **Edge-cloud trust orchestration:** Embedding EasyShield as a local pre-filter in Zero Trust architectures, enabling distributed biometric security with adaptive trust scoring.

EasyShield paves the way for practical, secure biometric authentication in embedded IoT, surveillance, and mobile applications by combining interpretability,

deployability, and strong detection margins, but it must be deployed as part of a layered security strategy.

Model Reproducibility and Source Code Availability.

The EasyShield v2.5 system was built upon a modified classification variant of the YOLOv12n architecture, originally released as part of the YOLOv12 project by SunSmartErJie [22]. The specific version used for our lightweight backbone is publicly available at <https://github.com/sunsmarterjie/yolov12/releases/tag/cls>. To support reproducibility and open research, the full implementation of EasyShield v2.5—including training pipeline, backbone modifications, and evaluation scripts—was developed and released by our collaborator Mohamed Wassim Mbarek [17].

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